VARUN KUMAR

Embedded Systems Engineer | C/C++ Firmware | RTOS | Driver Development

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PROFESSIONAL SUMMARY

Embedded engineer (B.Tech, Electronics and Computer Engineering) writing C/C++ firmware and device drivers for ARM Cortex-M microcontrollers (STM32, ESP32). Experienced in FreeRTOS task design, interrupt handling, and register-level drivers, and in hardware-firmware debugging with JTAG/SWD, OpenOCD, GDB, and logic analyzers. Has built containerized CI/CD pipelines for firmware validation. Eager to contribute modular C++ drivers, state-machine subsystem logic, and robust on-device software.

TECHNICAL SKILLS

Languages: C++ (OOP fundamentals), C, Embedded C, Python, Bash/Shell scripting, ARM Assembly

RTOS and Real-Time: FreeRTOS, task management, real-time scheduling, interrupt handling (NVIC, EXTI), state-based control logic, deterministic behavior, timing analysis

Microcontrollers: ARM Cortex-M (M0), STM32, ESP32, RISC-V, bare-metal programming, SDK integration, peripheral configuration

Drivers and Peripherals: GPIO, RCC, NVIC, EXTI, SYSCFG, ADC, SPI, I2C, UART, CAN; register-level programming, startup code, linker scripts

Communication and Networking: UART, SPI, I2C, CAN, TCP/IP, Bluetooth, RFID

Debugging and Test: JTAG/SWD, OpenOCD, GDB, oscilloscope, logic analyzer, multimeter, function generator, bench power supply; hardware-in-the-loop bring-up and validation

Build, CI/CD and Tools: GNU toolchain (GCC), Make, Git, GitHub, Docker, Jenkins CI/CD, SSH, Linux CLI

Embedded Linux: Fundamentals, device drivers, BSP development, kernel debugging

Hardware and IoT: Schematic design, PCB layout and routing, layer stackup, DFM, power management circuits, IoT prototyping, BMS systems

CAD: Fusion 360

PROFESSIONAL EXPERIENCE

Docker RISC-V Toolchain and CI/CD Infrastructure Development Intern | MindGrow, India *Apr 2026 – Jun 2026*

- Built a containerized embedded development environment for a RISC-V toolchain (OpenOCD, GDB, cross-compiler) in a single reproducible Docker image..
- Automated the firmware build and test pipeline with Jenkins CI/CD and a Docker prebuild step, enabling fast iteration and validation across multiple toolchain configurations.

NPD Engineer (Trainee) | TVS Sundram Fasteners Ltd, Tamil Nadu, India *Dec 2025 – Mar 2026*

- Supported new product development for industrial applications, from concept through prototype validation.
- Created product models and technical specifications in AutoCAD to ensure manufacturing precision and assembly accuracy.

Junior R&D Developer (Intern) | MH Cockpit, India *Jun 2025 – Oct 2025*

- Developed embedded firmware, circuit designs, and hardware-software integration for IoT devices and product prototypes.
- Performed hardware debugging, protocol analysis, embedded system testing, and prototype validation with lab equipment for product qualification.

PROJECTS

STM32F030x4 Peripheral Driver Library | C, ARM Cortex-M0, Makefile, OpenOCD | github.com/varunkumar898/Driver-for-STM32F030x4-x6-x8-xC

- Wrote bare-metal device drivers for GPIO, RCC, NVIC, EXTI, SYSCFG, ADC, SPI, and I2C on the STM32F030x4 (Cortex-M0, 16 KB flash, 4 KB SRAM) using register-level Embedded C, with no vendor HAL.
- Built the bare-metal toolchain: startup assembly, custom linker script, Makefile, ARM GCC, and OpenOCD for flashing and debugging.
- Maintained the project in Git with 17+ commits and public documentation covering features, pinout, and build steps.

Smart Number Plate for Theft Detection | C, FreeRTOS, ARM Cortex-M, RFID

- Developed bare-metal and FreeRTOS-based firmware on an ARM microcontroller for theft detection and control with real-time responsiveness.
- Implemented RFID protocol handling, closed-loop feedback control logic, and hardware protection circuits.
- Validated signal timing with a logic analyzer and debugged hardware-firmware integration issues using OpenOCD and GDB.

Embedded UI Designer (Final Year Project) | github.com/varunkumar898

- Built a desktop UI design tool for embedded systems with an SDK, a type-safe REST API engine, and multi-language firmware code generation for microcontrollers.

Opensource project (Gravel lancher – improve the preforms in the backend of the compliner LLVM)(**solidcraft –** i have port all the system function to arm device both window and arm)

EDUCATION

B.Tech, Electronics and Computer Engineering | Vels University, Chennai | CGPA: 7.5/10 *2026*